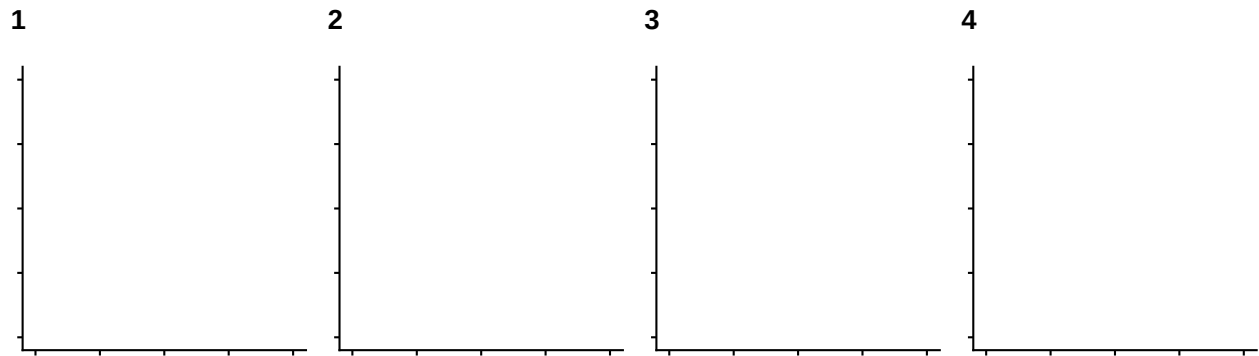


## Dice and dose-response relationships: toxins

Dose-response relationships frequently show up in biological (and other) systems. Indeed, I bet you can think of at least four *different* ways some *response* changes with the dose of some *cause*. Let's write those down:

- 1.
- 2.
- 3.
- 4.

In fact, draw graphs of those relationships. Add labels (including units if you can) and put reasonable numbers on the two axes.



It's good to keep this diversity in mind if only so that you don't make the mistake of assuming every dose-response relationship means the same thing.

## Dice determining susceptibility

To develop a more intuitive sense of at least one way we can get a dose-response relationship, let's roll some dice.

Imagine that each die represents some component of the mechanism(s) that help you remove, inactivate, repair damage from, or otherwise tolerate some toxin X<sup>1</sup>. For each die/component you might have a low (1) or high (6) value, or anything in between. The sum of values across all of your dice/mechanisms determines your *tolerance* of the toxin. That is, it determines how much exposure you can tolerate before you succumb.

<sup>1</sup> Imagine something specific if it helps, but it is not necessary for our little game.

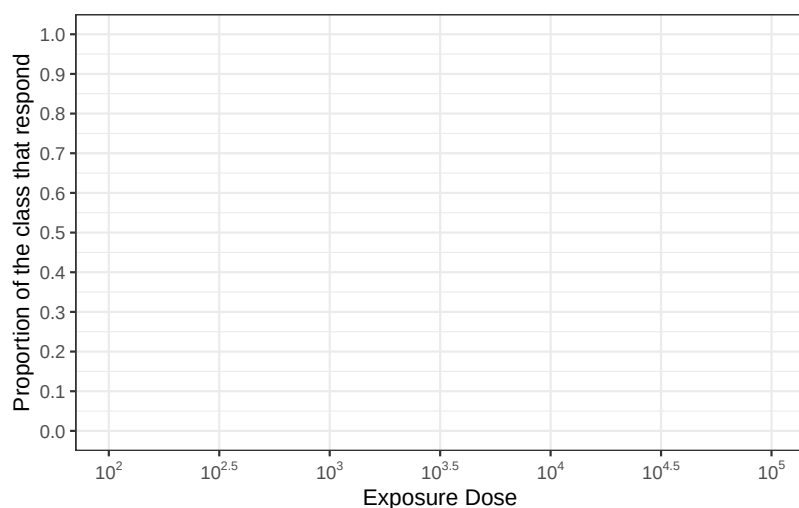
Here we go:

1. Please roll 10 dice and add up the numbers. Write that down here: \_\_\_\_\_
2. Divide that number by 10 and write it down here: \_\_\_\_\_ <-*This is your susceptibility*

## A dose-response experiment

At this point, every person in class should have their own (hidden) level of susceptibility. I, the experimenter, do not know anyone's susceptibility to toxin X, nor do even know anything useful like the average susceptibility of the class or the LD<sub>50</sub>. Let's see how I can use a dose-response study to figure out some of this.

Imagine I exposed the class to a particular dose of toxin X. I could see how many individuals (or the fraction of individuals) that *responded* to the exposure (i.e., died if that was our outcome of interest). That could then be plotted on the axes, below. The position along the x-axis is the dose and the position on the y-axis is the proportion of the class that responded.



Let's try this. I will expose the class to the dose and you simply raise your hand if you would respond to this dose. You should respond only if the *exponent* of the dose (i.e., the 2.5 in a dose  $10^{2.5}$ ) is *higher* than your tolerance (the number you wrote above).

| Dose                | $10^2$ | $10^{2.5}$ | $10^3$ | $10^{3.5}$ | $10^4$ | $10^{4.5}$ | $10^5$ |
|---------------------|--------|------------|--------|------------|--------|------------|--------|
| Number of responses |        |            |        |            |        |            |        |

Now let's plot these data.

Do they look like what you expected? What would happen if we repeated this “experiment”? Would we get the same or similar results? Why?

## Useful and not

As always, our little toy model is useful in some ways and not-so-useful in other ways. Think about what was useful for helpful about this activity and what is missing, inaccurate, misleading, or otherwise not so helpful.